PROGRESS PBL FASE 1

Kelompok 6



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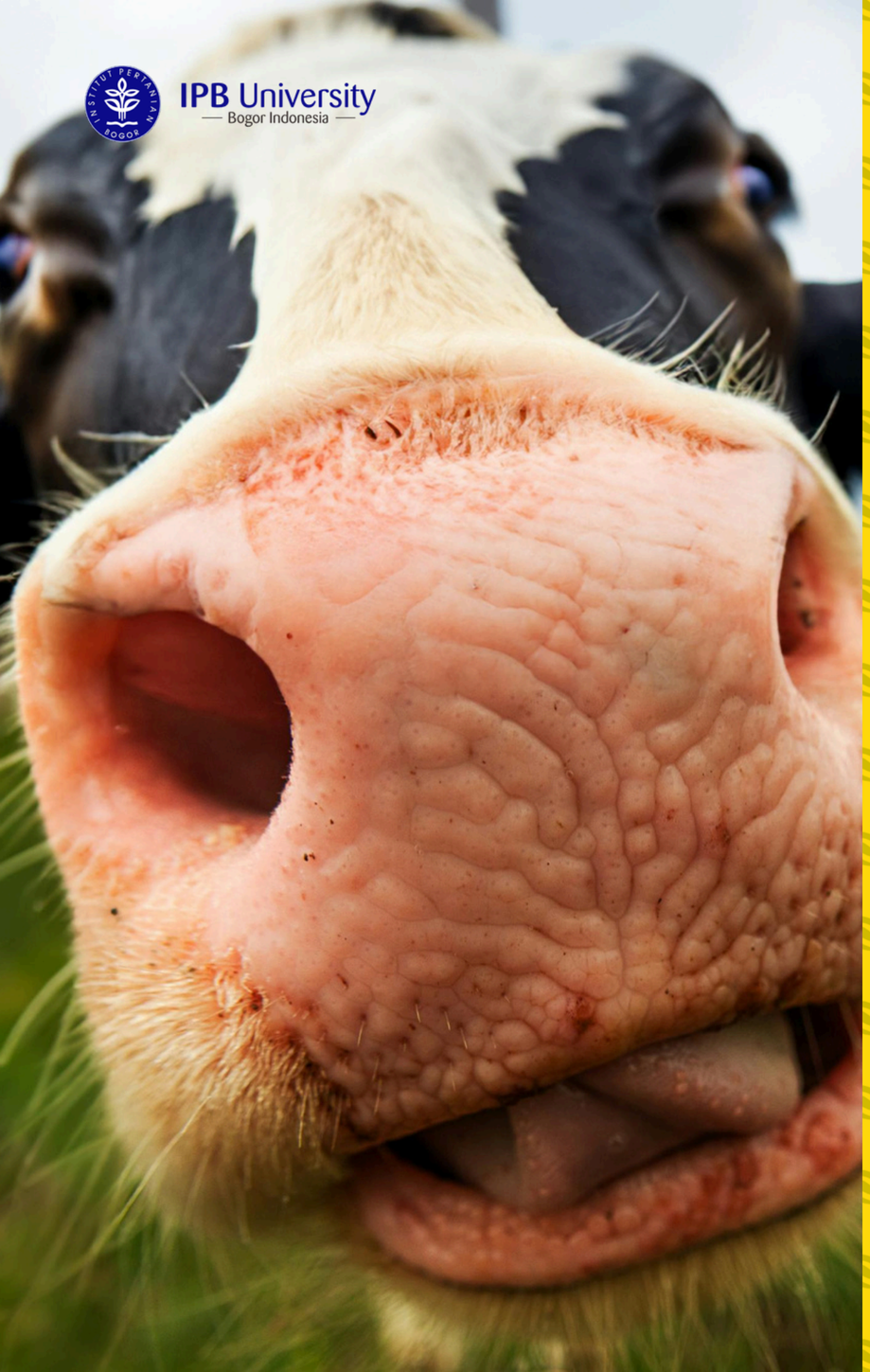
Anggota Kelompok

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Lead Analyst

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Red Team

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Blue Team

Yuwandri Alfarizi (J0404241010)
Blue Team



Deskripsi Projek IoT

- Tujuan: Mengembangkan sistem pemantauan dan pengendalian heat stress pada sapi secara otomatis.
- Monitoring IoT: Mengukur suhu dan kelembapan kandang untuk menghitung Temperature-Humidity Index (THI).
- Computer Vision: Menganalisis perilaku sapi melalui CCTV, khususnya durasi berdiri dan berbaring.
- Analisis Data: Mengidentifikasi hubungan antara kondisi mikroklimat kandang, tingkat THI, dan perubahan perilaku sapi.
- Automated Cooling: Merancang sistem penyiraman otomatis pada periode kritis untuk mengurangi risiko heat stress.
- Dashboard Monitoring: Menampilkan data lingkungan, perilaku sapi, dan status sistem secara terintegrasi.

1. Giannone et al. (2023), Review of the Heat Stress-Induced Responses in Dairy Cattle. Membahas respons sapi terhadap heat stress, indikator fisiologis, dan THI. https://doi.org/10.3390/ani13223451
2. Tucker et al. (2021), Invited review: Lying time and the welfare of dairy cows. Membahas durasi, frekuensi, dan pola berbaring sapi serta kaitannya dengan kesejahteraan hewan. https://doi.org/10.3168/jds.2019-18074
3. Asmarasari et al. (2023), A review of dairy cattle heat stress mitigation in Indonesia. Membahas kondisi heat stress sapi perah di Indonesia dan strategi mitigasinya. https://doi.org/10.14202/vetworld.2023.1098-1108

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Latar Belakang Produk IoT

LATAR BELAKANG

• Suhu dan kelembapan tinggi dapat meningkatkan risiko heat stress pada sapi.

• Respons panas berbeda antar sapi; data lingkungan perlu dilengkapi indikator berbasis hewan.

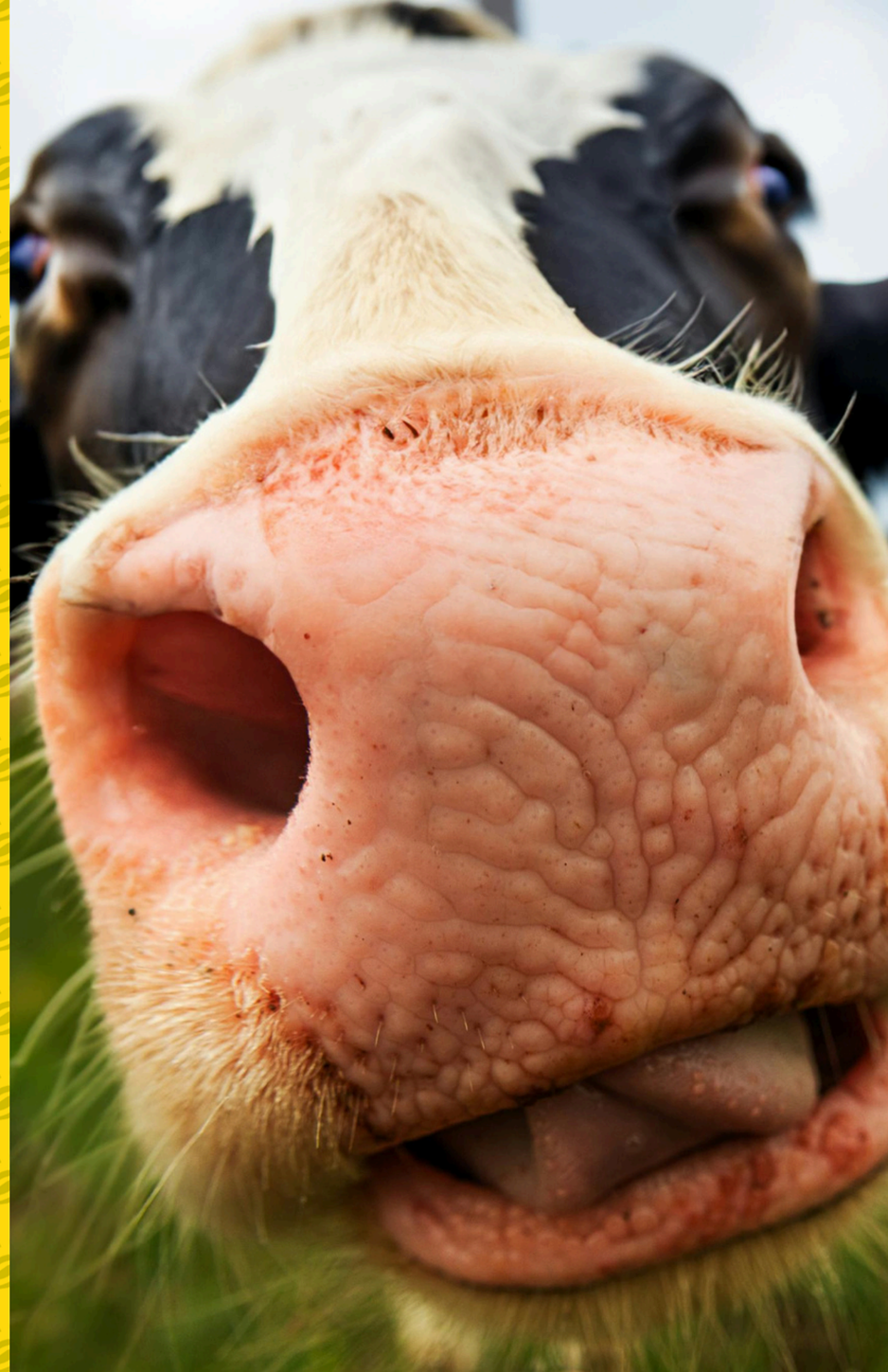
• Diperlukan pemantauan terintegrasi sebagai dasar peringatan dan tindakan pendinginan.

RUMUSAN MASALAH

1. Apakah terdapat hubungan antara peningkatan THI dengan perubahan durasi berdiri dan berbaring pada sapi?

2. Apakah penyiraman terkontrol pada periode kritis memperbaiki kondisi termal dan perilaku istirahat sapi, tanpa membuat area istirahat terlalu basah?

3. Bagaimana perubahan kondisi lingkungan dan perilaku sapi setelah dilakukan pendinginan terkontrol?



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Komponen	Proyek IoT Acuan	Target Node
Sistem operasi	Ubuntu Server	Ubuntu Server
Database	Firebase	SQLite
Komunikasi sensor	HTTP REST/JSON	HTTP REST pada port 80
Komunikasi kamera	RTSP	RTSP pada port 554
Dashboard	Web Node.js	Web Node.js pada port 80
Akses administrasi	SSH key-based dan web admin	SSH port 22
Monitoring keamanan	Syslog ke collector terpusat	Syslog ke Security Onion (192.168.6.30)

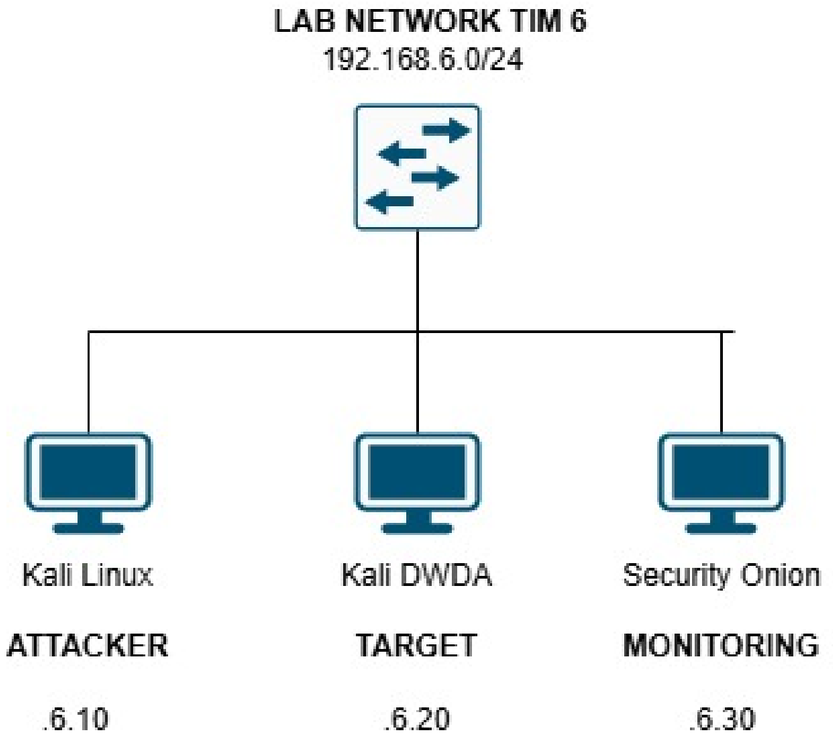
Pemetaan Teknologi

Mereplikasi teknologi backend asli ke Target Node Virtual Machine lab untuk menjamin risiko nol pada sistem yang dibuat

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Arsitektur Jaringan & Skema IP

Hostname	IP Address	OS	Peran
attacker-kel6	192.168.6.10	Kali Linux 2026.2	Attacker / Red Team
target-kel6	192.168.6.20	Ubuntu Server 26.04.1 LTS	Target / korban
monitor-kel6	192.168.6.30	Security Onion	Monitoring / Blue Team



Progres Hardening Target Node

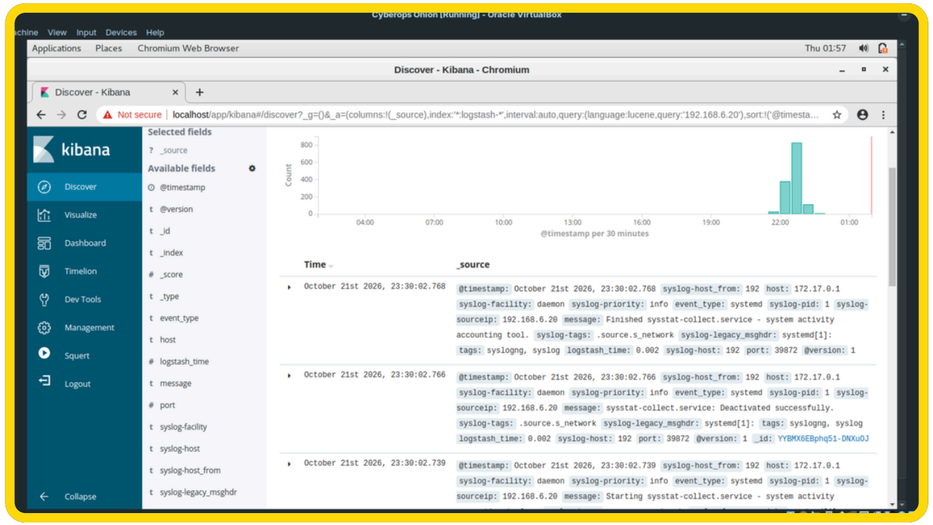
17	15	2	3
kontrol keamanan	selesai	sebagian atau belum	port terbuka

KONTROL KUNCI

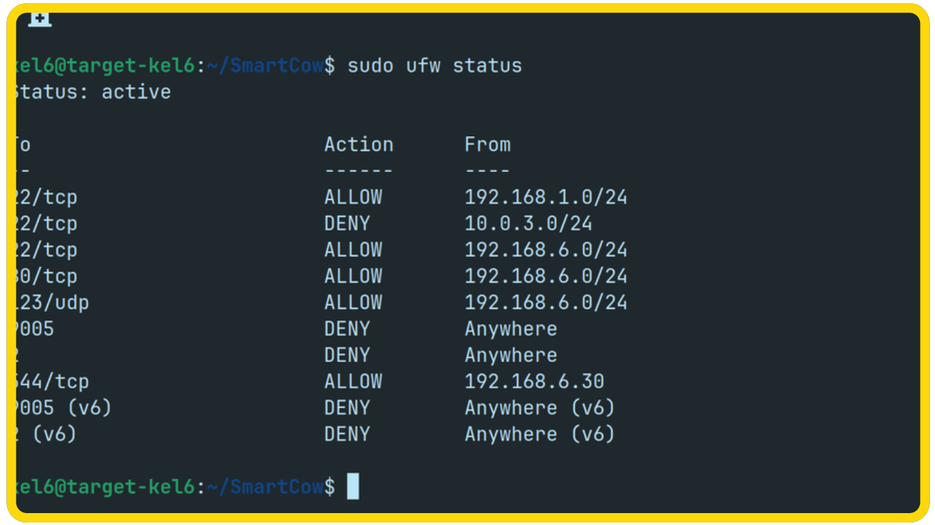
Akses	SSH key-based, root login off, MaxAuthTries 3, hanya user kel6.
Permukaan serangan	UFW default deny, port 22/80/123 hanya untuk /24, RTSP 554 hanya .30.
Layanan	Protokol MediaMTX yang tidak dipakai dimatikan, OS dan paket dipatch.
Data	SQLite dan config berizin 640, database tidak bisa diunduh lewat HTTP.
Aplikasi	Password scrypt, session 12 jam, rate limit login 10 percobaan per menit.

Bukti before/after tersimpan di docs/blueteam/evidence. Yang masih sebagian: daftar ban fail2ban dan penggantian password awal akun kel6.

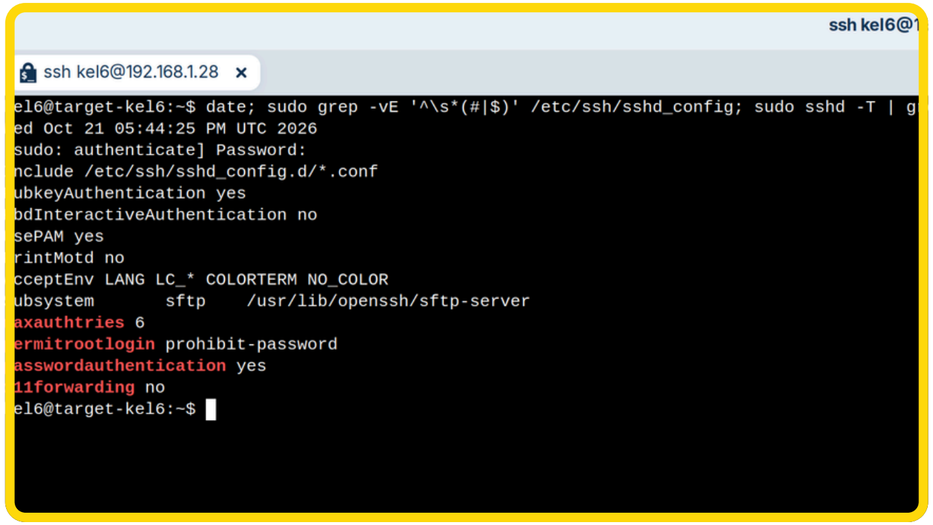
Bukti Hardening



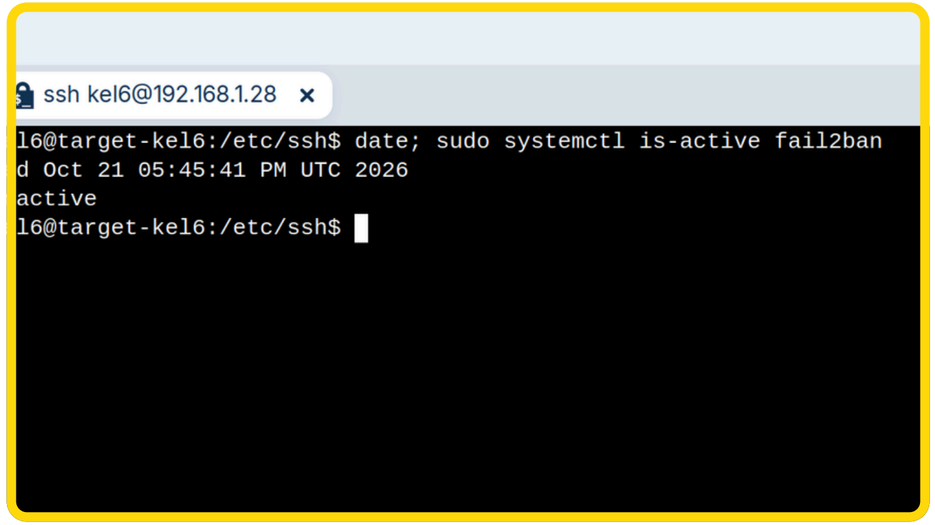
Monitoring Baseline Log



Salah Konfigurasi UFW 544



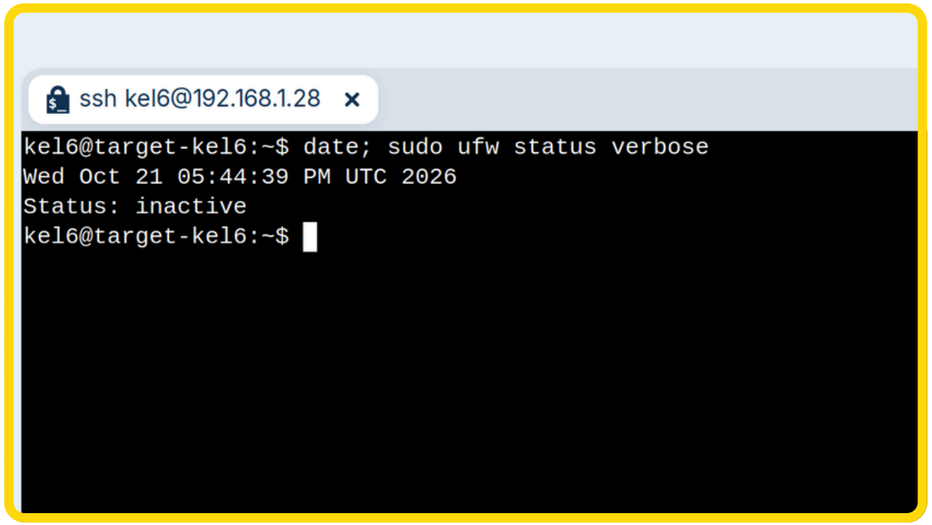
Sebelum 1 SSH



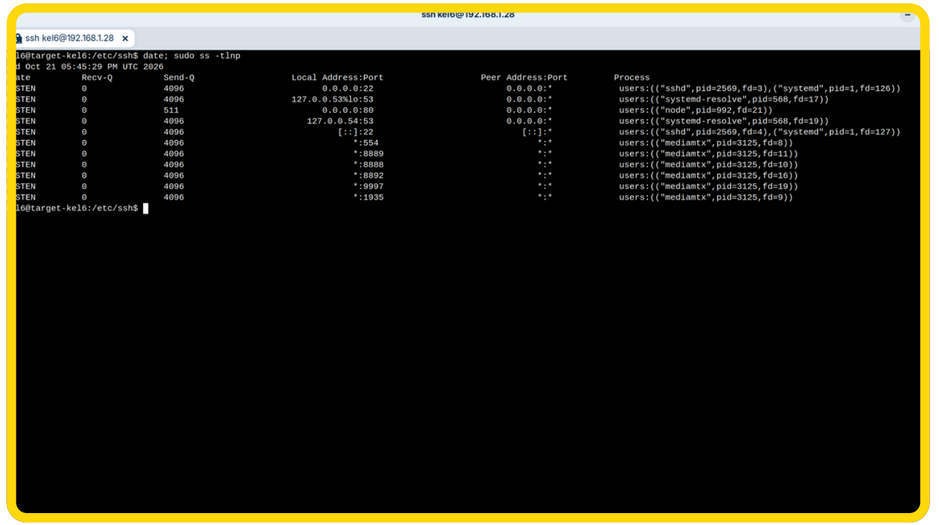
Sebelum 2 Fail 2 Ban

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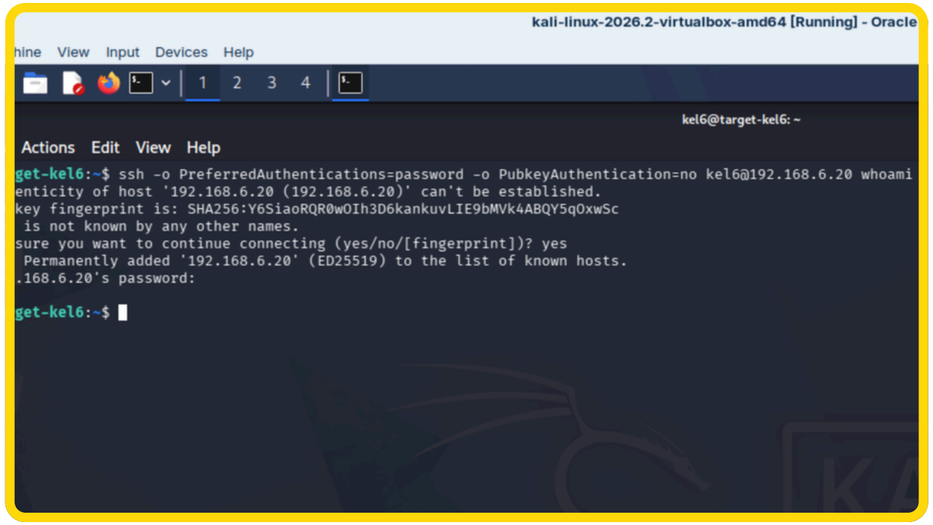
Bukti Hardening



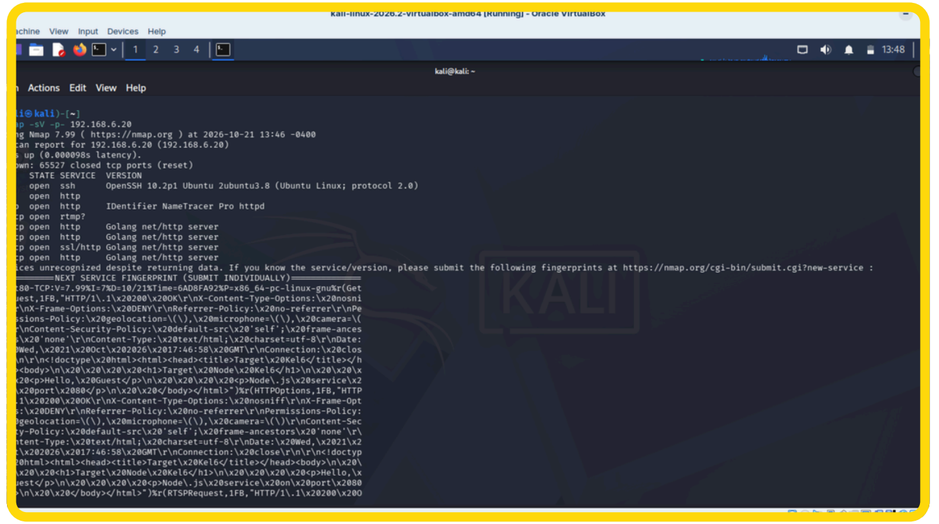
Sebelum 3 Firewall



Sebelum 4 Port



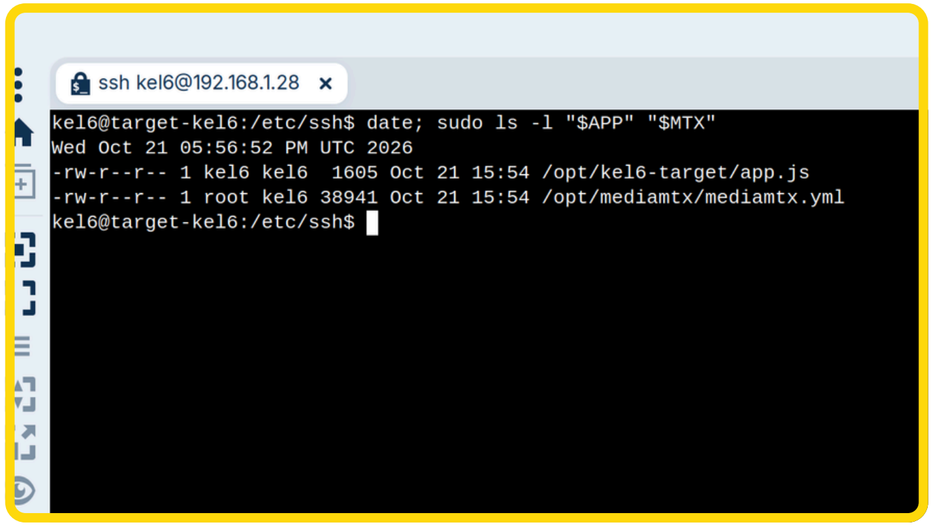
Sebelum 5 Login



Sebelum 5 Pindai NMap

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Bukti Hardening



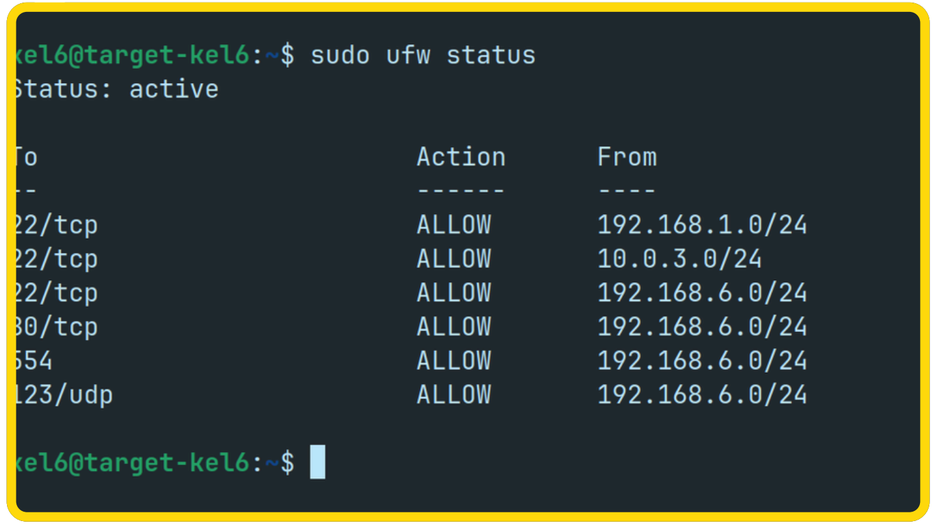
Sebelum 7 Izin Berkas



Sesudah 1 a Konfigurasi SSH



Sesudah 2a Konfigurasi Fail2ban



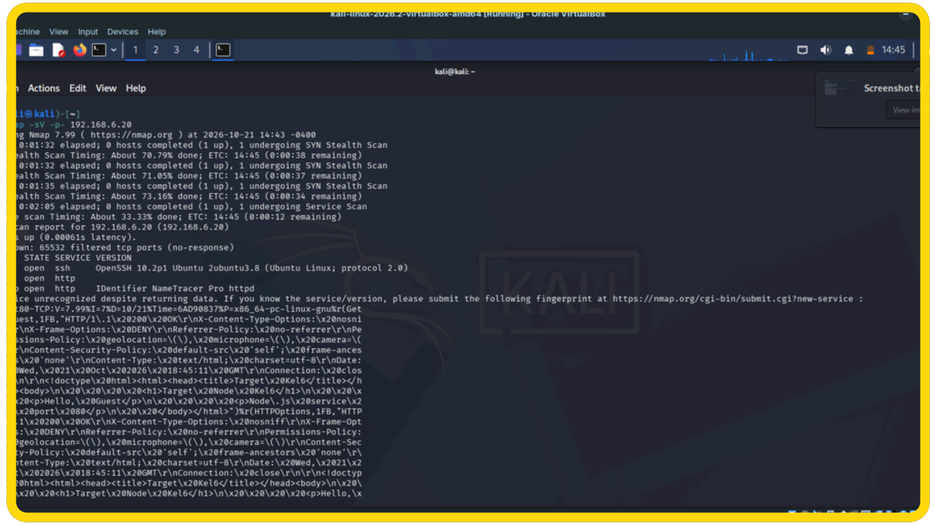
Sesudah 3a Aturan Firewall

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Bukti Hardening

```
kel6@target-kel6:~$ sudo grep -nE 'rtspAddress|rtmpAddress|h
149:apiAddress: :9997
173:metricsAddress: :9998
221:playbackAddress: :9996
250:rtspAddress: :554
295:rtmpAddress: :1935
316:hlsAddress: :8888
379:webrtcAddress: :8889
439:srtAddress: :8890
kel6@target-kel6:~$
```

Sesudah 4a Konfigurasi Mediamtx



Sesudah 5 Pindai NMap

```
kel6@target-kel6:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    192.168.1.0/24
22/tcp                     ALLOW IN    10.0.3.0/24
22/tcp                     ALLOW IN    192.168.6.0/24
80/tcp                     ALLOW IN    192.168.6.0/24
554                        ALLOW IN    192.168.6.0/24
123/udp                    ALLOW IN    192.168.6.0/24

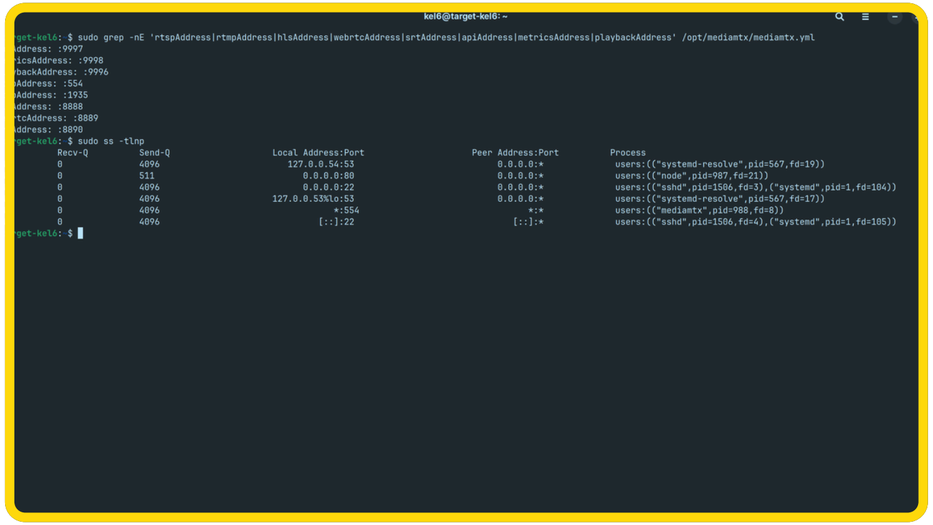
kel6@target-kel6:~$
```

Sesudah 6 Login Ditolak

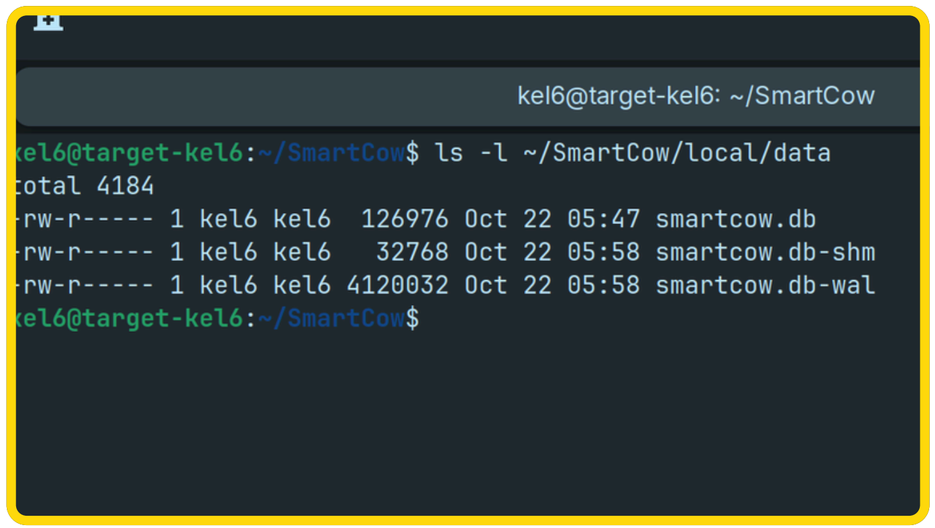
```
kel6@target-kel6:~$ sudo grep -nE 'rtspAddress|rtmpAddress|h
149:apiAddress: :9997
173:metricsAddress: :9998
221:playbackAddress: :9996
250:rtspAddress: :554
295:rtmpAddress: :1935
316:hlsAddress: :8888
379:webrtcAddress: :8889
439:srtAddress: :8890
kel6@target-kel6:~$
```

Sesudah 7a Izin Berkas

Bukti Hardening



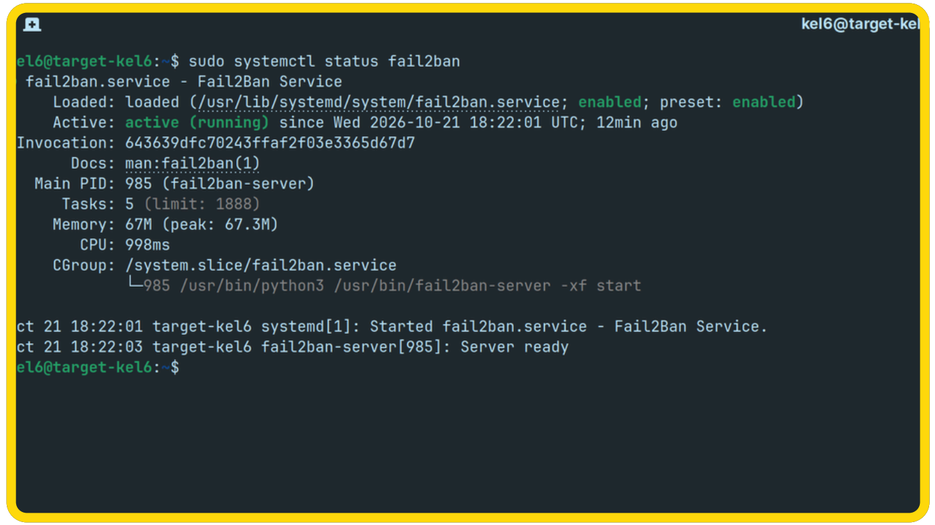
Sesudah 9a Brute Force



Sesudah 18a Izin DB



Sesudah 1b SSH Efektif



Sesudah 2b Konfigurasi Fail2ban

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Bukti Hardening

```
kel6@target-kel6:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    192.168.1.0/24
22/tcp                     ALLOW IN    10.0.3.0/24
22/tcp                     ALLOW IN    192.168.6.0/24
80/tcp                     ALLOW IN    192.168.6.0/24
554                        ALLOW IN    192.168.6.0/24
123/udp                    ALLOW IN    192.168.6.0/24

kel6@target-kel6:~$
```

Sesudah 3b Status Firewall

```
kel6@target-kel6: ~
get-kel6: $ sudo grep -nE 'rtspAddress|rtmpAddress|hlsAddress|webrtcAddress|srtAddress|apiAddress|metricsAddress|playbackAddress' /opt/mediamtx/mediamtx.yml
ddress: :9997
icsAddress: :9998
backAddress: :9996
Address: :554
Address: :1935
ddress: :8888
rtcAddress: :8889
ddress: :8890
get-kel6: $ sudo ss -tlnp
        Recv-Q          Send-Q                    Local Address:Port                    Peer Address:Port         Process
        0               4096                      127.0.0.54:53                          0.0.0.0:*             users:(("systemd-resolve",pid=567,fd=19))
        0               511                         0.0.0.0:80                           0.0.0.0:*             users:(("node",pid=987,fd=21))
        0               4096                        0.0.0.0:22                           0.0.0.0:*             users:(("sshd",pid=1586,fd=3),("systemd",pid=1,fd=104))
        0               4096                   127.0.0.53%lo:53                          0.0.0.0:*             users:(("systemd-resolve",pid=567,fd=17))
        0               4096                           *:554                              *:*                 users:(("mediamtx",pid=988,fd=8))
        0               4096                           [::]:22                         [::]:*                 users:(("sshd",pid=1586,fd=4),("systemd",pid=1,fd=105))
get-kel6: $
```

Sesudah 4b Port Listen

```
kel6@target-kel6: ~
get-kel6: $ sudo grep -nE 'rtspAddress|rtmpAddress|hlsAddress|webrtcAddress|srtAddress|apiAddress|metricsAddress|playbackAddress' /opt/mediamtx/mediamtx.yml
ddress: :9997
icsAddress: :9998
backAddress: :9996
Address: :554
Address: :1935
ddress: :8888
rtcAddress: :8889
ddress: :8890
get-kel6: $ sudo ss -tlnp
        Recv-Q          Send-Q                    Local Address:Port                    Peer Address:Port         Process
        0               4096                      127.0.0.54:53                          0.0.0.0:*             users:(("systemd-resolve",pid=567,fd=19))
        0               511                         0.0.0.0:80                           0.0.0.0:*             users:(("node",pid=987,fd=21))
        0               4096                        0.0.0.0:22                           0.0.0.0:*             users:(("sshd",pid=1586,fd=3),("systemd",pid=1,fd=104))
        0               4096                   127.0.0.53%lo:53                          0.0.0.0:*             users:(("systemd-resolve",pid=567,fd=17))
        0               4096                           *:554                              *:*                 users:(("mediamtx",pid=988,fd=8))
        0               4096                           [::]:22                         [::]:*                 users:(("sshd",pid=1586,fd=4),("systemd",pid=1,fd=105))
get-kel6: $
```

Sesudah 8b Tidak ada Pembaruan

```
kali@kali: ~
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ for p in \
  /local/data/smartcow.db \
  /local/data/smartcow.db-wal \
  /data/smartcow.db \
  /js/../local/data/smartcow.db \
  /js/%2e%2e/local/data/smartcow.db
do
  code=$(curl -s -o /dev/null -w '%{http_code}' --path-as-is "http://192.168.6.20$p")
  printf '%-45s %s\n' "$p" "$code"
done
/local/data/smartcow.db                       404
/local/data/smartcow.db-wal                   404
/data/smartcow.db                             404
/js/../local/data/smartcow.db                 404
/js/%2e%2e/local/data/smartcow.db             404

┌──(kali㉿kali)-[~]
└─$ curl -I 192.168.6.20/local/data/smartcow.db
HTTP/1.1 404 Not Found
Content-Type: application/json; charset=utf-8
Cache-Control: no-store
X-Content-Type-Options: nosniff
Date: Thu, 22 Oct 2026 06:03:14 GMT
Connection: keep-alive
Keep-Alive: timeout=5
```

Sesudah 18b DB 404

```
kali@kali: ~
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ for i in $(seq 1 12); do
  code=$(curl -s -o /tmp/rl.json -w '%{http_code}' \
    -X POST http://192.168.6.20/api/login \
    -H 'Content-Type: application/json' \
    -d '{"email":"admin@smartcow.local","password":"salah"}')
  printf '%2d  %s  %s\n' "$i" "$code" "$(cat /tmp/rl.json)"
done
 1  429  {"error":"Terlalu banyak percobaan. Coba lagi sebentar."}
 2  429  {"error":"Terlalu banyak percobaan. Coba lagi sebentar."}
 3  429  {"error":"Terlalu banyak percobaan. Coba lagi sebentar."}
 4  429  {"error":"Terlalu banyak percobaan. Coba lagi sebentar."}
 5  429  {"error":"Terlalu banyak percobaan. Coba lagi sebentar."}
 6  429  {"error":"Terlalu banyak percobaan. Coba lagi sebentar."}
 7  429  {"error":"Terlalu banyak percobaan. Coba lagi sebentar."}
 8  429  {"error":"Terlalu banyak percobaan. Coba lagi sebentar."}
 9  429  {"error":"Terlalu banyak percobaan. Coba lagi sebentar."}
10  429  {"error":"Terlalu banyak percobaan. Coba lagi sebentar."}
11  429  {"error":"Terlalu banyak percobaan. Coba lagi sebentar."}
12  429  {"error":"Terlalu banyak percobaan. Coba lagi sebentar."}

┌──(kali㉿kali)-[~]
```

Sesudah 19b Rate Limit

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Rencana Skenario Serangan

ID	Skenario Serangan	Target	Metode / Tools	Tujuan Serangan	Expected Defense
1	Port Scanning	Server 192.168.6.20	nmap	Mengidentifikasi port dan service yang terbuka	Mengidentifikasi port dan service yang terbuka
2	Unauthorized Web Access	Website / endpoint monitoring	curl, browser, Burp Suite	Menguji apakah dashboard atau endpoint bisa diakses tanpa login	Sistem meminta autentikasi / redirect ke login / 401 atau 403
3	Login Brute Force Simulation	Halaman login web	Burp Suite / request berulang	Menguji batas percobaan login	Rate limiting aktif, respons 429 Too Many Requests
4	Session Hijacking Simulation	Session login website	Browser DevTools / Burp Suite	Menguji apakah session lama masih bisa dipakai setelah logout / expire	Session invalid setelah logout atau expired
5	Database File Exposure	File database SQLite	curl, browser	Menguji apakah file database dapat diakses melalui web	File database tidak bisa diakses, respons 403 / 404
6	Unauthorized RTSP Access	RTSP stream / MediaMTX	VLC, ffprobe	Menguji apakah stream CCTV bisa diakses tanpa izin	Akses RTSP ditolak jika tanpa kredensial / IP tidak diizinkan

TERIMA KASIH